Stage 2 — Check Your Understanding

Q1. Where the pressure comes from (Step 1, §4.2)

Starting equations (4.1): $\mu\nabla^2 u - \nabla p + f\,\varphi_\varepsilon = 0$, with $\nabla\cdot u = 0$ and constant force vector f.

1. Take divergence (linearity): $\mu\nabla\cdot(\nabla^2 u) - \nabla\cdot(\nabla p) + \nabla\cdot(f\,\varphi_\varepsilon) = 0$.

2. Write in index notation: $\nabla\cdot(\nabla^2 u) = \partial_i(\partial_j\partial_j u_i)$.

3. Commute partial derivatives (smooth field): $\partial_i\partial_j\partial_j u_i = \partial_j\partial_j\partial_i u_i$.

4. Use incompressibility here: $\nabla\cdot(\nabla^2 u) = \nabla^2(\nabla\cdot u) = \nabla^2(0) = 0$.

5. Pressure term: $\nabla\cdot(\nabla p) = \nabla^2 p$; force term: $\nabla\cdot(f\,\varphi_\varepsilon) = f\cdot\nabla\varphi_\varepsilon$ because f is constant.

6. Hence $-\nabla^2 p + f\cdot\nabla\varphi_\varepsilon = 0$, so $\nabla^2 p = f\cdot\nabla\varphi_\varepsilon$.

7. If $\nabla^2 G = \varphi_\varepsilon$, then $p = f\cdot\nabla G$ is a particular solution, up to an additive constant and admissible harmonic contribution.

Q2. Checking G (Step 3)

Let $X = r^2/\varepsilon^2$ and $G(r) = [\ln(r^2) + E_1(X)]/(4\pi)$.

1. Chain-rule inner derivative: $dX/dr = 2r/\varepsilon^2$.

2. Exponential integral derivative: $dE_1/dX = -\exp(-X)/X$.

3. Chain rule: $dE_1(r^2/\varepsilon^2)/dr = [-\exp(-X)/X](2r/\varepsilon^2) = -2\exp(-X)/r$.

4. First radial derivative: $G'(r) = [2/r - 2\exp(-X)/r]/(4\pi) = [1 - \exp(-X)]/(2\pi r)$.

5. Multiply: $rG' = [1 - \exp(-X)]/(2\pi)$.

6. Differentiate: $(rG')' = \exp(-X)(2r/\varepsilon^2)/(2\pi) = r\exp(-X)/(\pi\varepsilon^2)$.

7. Divide by r: $(1/r)(rG')' = \exp(-r^2/\varepsilon^2)/(\pi\varepsilon^2) = \varphi_\varepsilon(r)$.

The identity holds for $r > 0$ and extends continuously to $r = 0$.

Q3. Proposition 1 in your own words

For a closed fibre boundary C enclosing D, the divergence theorem gives $\oint_C u\cdot n\,ds = \int_D \nabla\cdot u\,dA = 0$. Gaussian regularization makes the velocity smooth throughout D, including the nominal fibre interior, so the theorem applies without a singularity exception. Fluid entering the fibre contributes negative outward flux; fluid leaving contributes positive outward flux, and the two cancel.

Table 2: Leakage measures as fractions of 2aU.

N	ε/h	Net flux	Q_+	Qc
8	0.06	$< 10^{-15}$	0.305	0.305
32	0.25	$< 10^{-15}$	0.0284	0.0284
32	1.02	$< 10^{-15}$	1.08×10^{-6}	1.28×10^{-6}

Requested two sentences: The positive-part flux Q_+ integrates only outward velocity, so inward and outward leakage cannot cancel. The chord flux Qc measures flow crossing the fibre diameter and therefore exposes through-flow despite zero net flux.

Q4. Why one blob gets Hasimoto right

For a cylinder of radius a, combine the 2D Stokeslet and a potential dipole of strength $a^2/2$:

$$4\pi\mu u = (-\ln r - 1/2)f + (f\cdot x)x/r^2 + (a^2/2)[f/r^2 - 2(f\cdot x)x/r^4].$$

1. Substitute r = a:

$$4\pi\mu u(a) = (-\ln a - 1/2)f + (f\cdot x)x/a^2 + f/2 - (f\cdot x)x/a^2.$$

2. Both direction-dependent terms cancel; $-f/2$ and $+f/2$ also cancel. Therefore $4\pi\mu u(a) = -f \ln a$, constant around the cylinder surface.

3. At the Gaussian blob centre, equation (4.11) gives $4\pi\mu u(0) = (-\ln \varepsilon + \gamma/2)f$. Equating to $-f \ln a$ yields $\ln \varepsilon = \ln a + \gamma/2$, or $\varepsilon = a \exp(\gamma/2) \approx 1.3346a$. Thus $a_{eff} = \varepsilon \exp(-\gamma/2) \approx 0.749\varepsilon$.

4. The matched blob has far-field dipole strength $\varepsilon^2/2$, versus $a^2/2$ for the physical cylinder. Their ratio is $\varepsilon^2/a^2 = \exp(\gamma) \approx 1.7811$, so the blob dipole is approximately 78.1% stronger.

5. This dipole difference affects the first concentration correction: the square-array cylinder series has $+2\alpha$ inside the permeability bracket, whereas the blob produces approximately $+1.78\alpha$. The discrepancy therefore becomes more important as α and fibre interactions increase.

6. Figure 5(a) and Proposition 3 report the dilute Hasimoto constant 1.4762 at $\alpha = 10^{-4}$, and at $\alpha = 0.1$ the constants 1.2982 (blob) and 1.2897 (cylinder), corresponding to about 0.8% permeability difference.